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HIGH VOLTAGE ELECTRICAL SYSTEM DESIGN

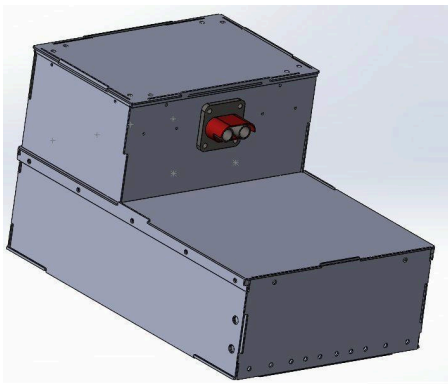
Sydney Motorsport | Electrical Department Head | 2024 – 2026

[470V Battery System](#) | [PCB Design \(Fusion360\)](#) | [LTspice](#) | [SolidWorks](#)

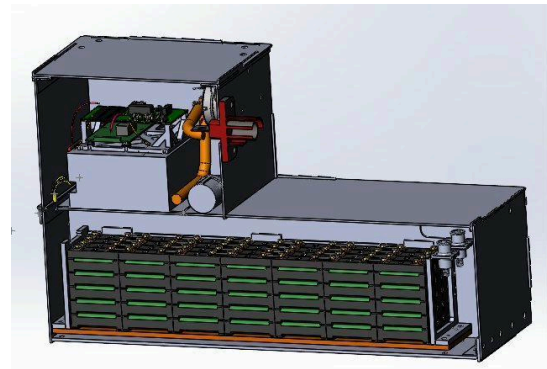
Led the high voltage subsystem design for Sydney Motorsport's first electric vehicle, encompassing the full accumulator, inverter integration, motor systems, and LV/HV interface. As former Electrical Department Head and High Voltage Lead, responsible for the complete electrical architecture from concept through validation.

Key System Components:

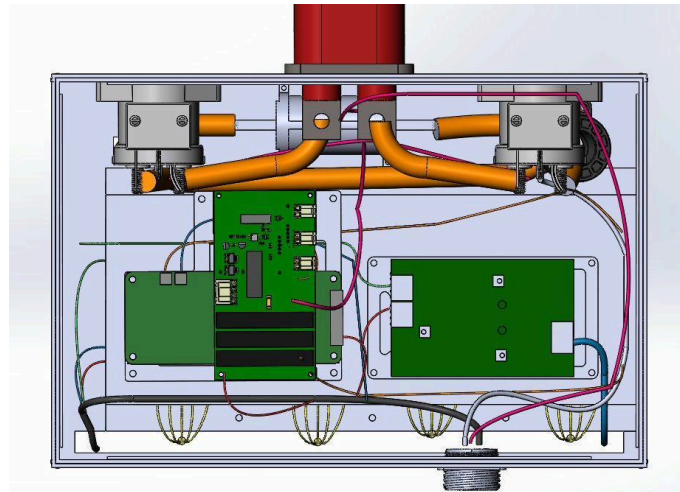
- **470.4V Battery Pack:** 112 ENEPAQ 18650VCT5A modules (1s5p configuration)
- **Isolation Relays (IRs):** Isolate battery until vehicle is ready
- **BMS, IMD, Precharge, TEMs:** Full monitoring, safety, and startup control



Accumulator Enclosure (CAD)



Section View — Modules & Electronics



LV control electronics compartment — PCBs, isolation relays, and wiring harness

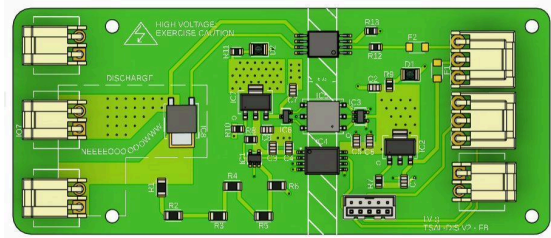
TSAL-DISCHARGE CIRCUIT

Fusion360 | LTspice | PCB Layout | Comparator Networks

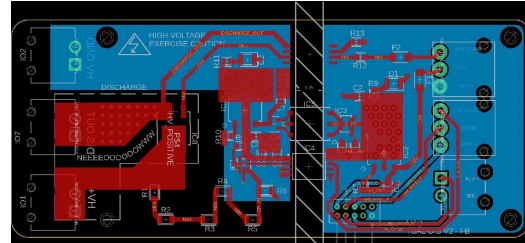
Combined TSAL and discharge circuit serving two critical safety functions: indicating when the tractive system voltage exceeds 60V via a visible light on the roll-hoop, and safely discharging the tractive system capacitors when the vehicle shuts down. This single board replaced two separate circuits, simplifying wiring and improving maintainability.

Design: Voltage-sensing comparator network with 60V reference threshold. When shutdown signal goes LOW, the discharge circuit creates a closed RC path through the inverter capacitive load.

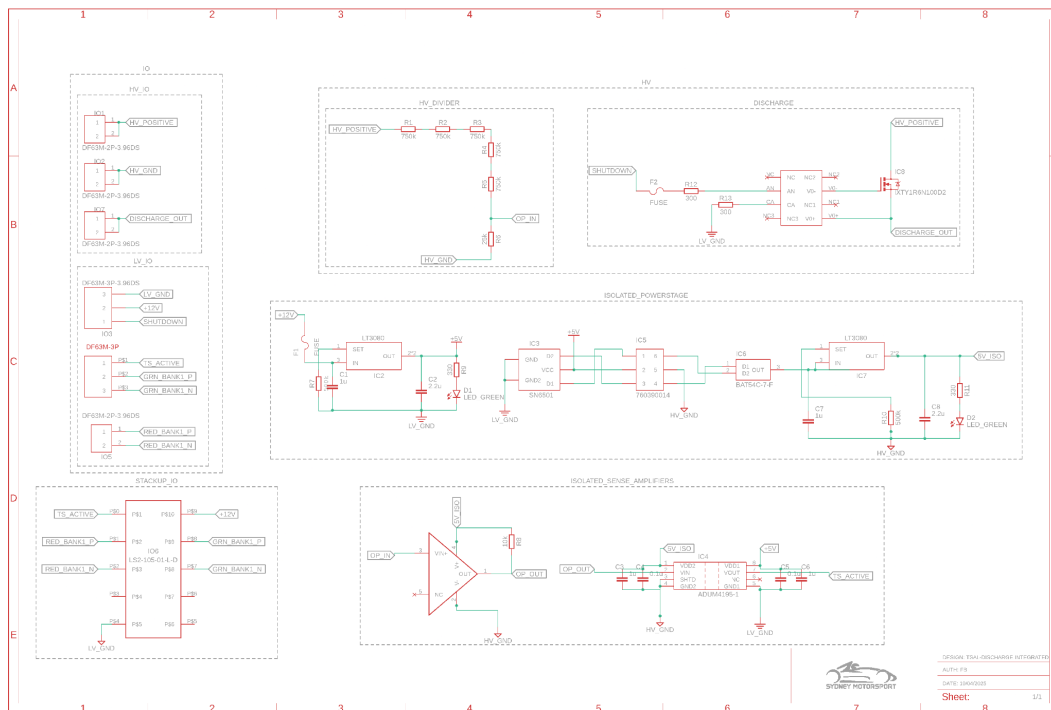
Result: Successfully validated and tested in the full electrical system. Currently integrated in the 2025 EV build.



TSAL-Discharge PCB (3D Render)



PCB Layout (Fusion360)



TSAL-DIS Schematic (Fusion360)

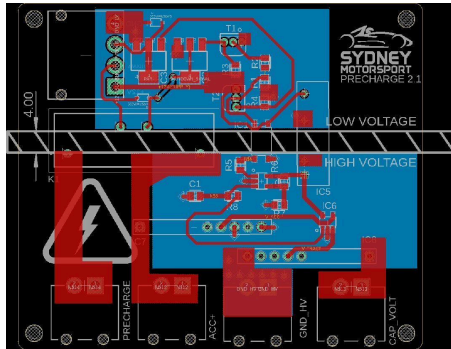
PRECHARGE CIRCUIT

Fusion360 | LTspice | Relay Control | Voltage Sensing

Safety-critical subsystem managing the initial connection between the accumulator and inverter. Limits inrush current during startup and ensures tractive system voltage rises smoothly to >95% of accumulator voltage before the positive isolation relay closes.

Design: When shutdown signal is HIGH, the precharge relay closes to connect the accumulator through a current-limiting resistor. A voltage sensing network detects when the tractive system reaches >95% of max voltage, then closes the Positive Isolation Relay to deliver full power.

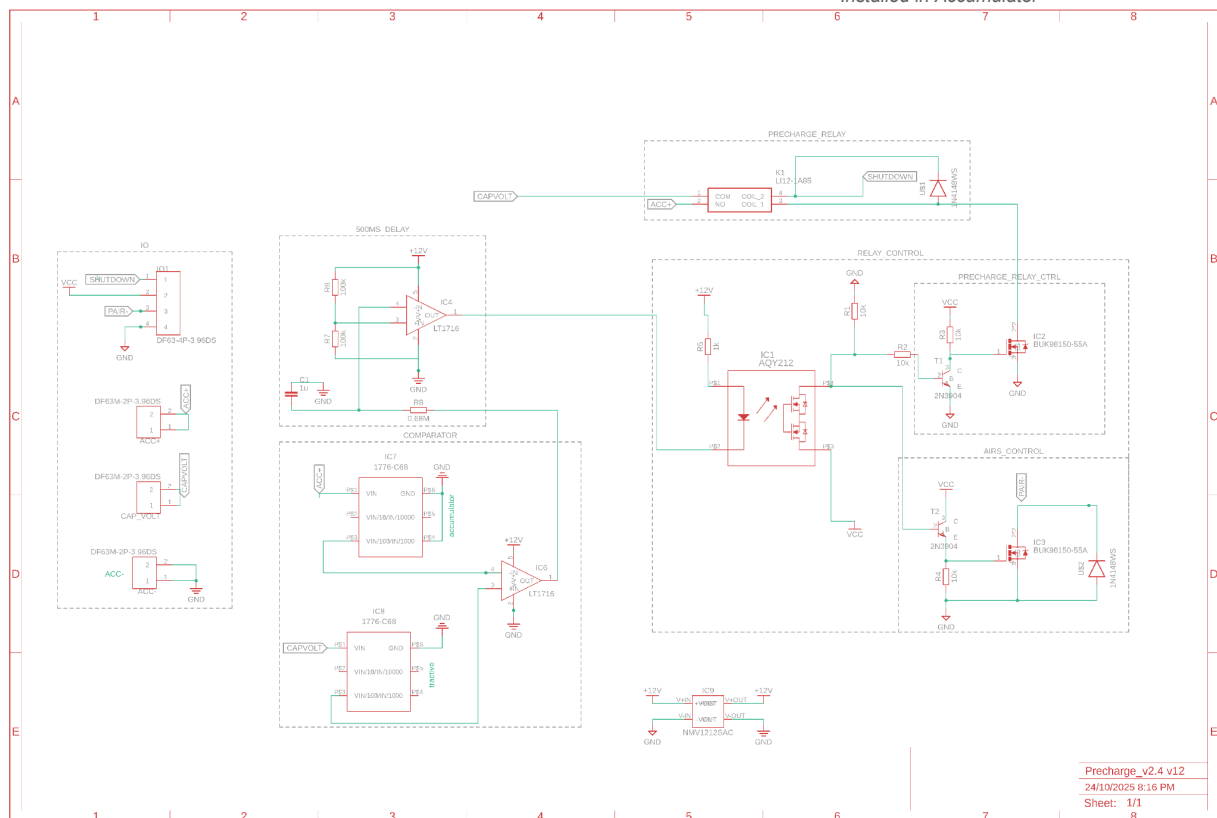
Result: Achieved 95% charge in <5 seconds as required by regulations. Currently in use in the 2025 EV build.



PCB Layout (Fusion360)



Installed in Accumulator



Precharge Schematic (Fusion360)

Precharge_v2.4 v12
24/10/2025 8:16 PM
Sheet: 1/1

UNIVERSITY PROJECTS

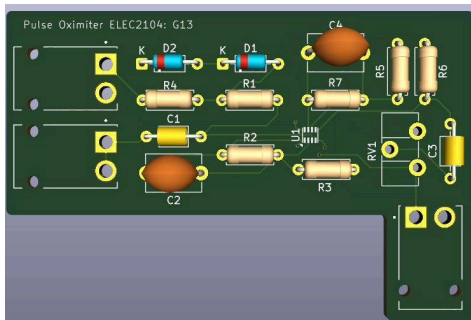
PULSE OXIMETER CIRCUIT

ELEC2104: Electronic Devices and Circuits

[KiCad](#) | [Analog Design](#) | [RC Filtering](#) | [BJT Phototransistor](#) | [Oscilloscope Validation](#)

Fully analog pulse oximeter circuit measuring heartbeat and SpO2 using IR and red LEDs passing light through to a BJT phototransistor. Sensor output passes through a two-stage cascading RC filter network designed to isolate the 1–3 Hz frequency range corresponding to the human heart rate, attenuating ambient light noise and high-frequency interference.

Result: Successfully detected pulse waveforms between 1–2 Hz. Two-stage RC filter effectively suppressed noise, producing a stable and repeatable signal validated across multiple users. Successfully detected SpO2 and Heart Rate using oscilloscope measurements. Received 98% on the project.



PCB 3D Render (KiCad)



Output — Red LED On

BEAT-SYNC VFX

MTRX3700: Mechatronics 3 | FPGA Group Project

[Verilog/SystemVerilog](#) | [FPGA \(DE2-115\)](#) | [VGA Controller](#) | [5×5 Convolution](#) | [DSP](#) | [ModelSim](#)

FPGA-based system processing live audio input through a microphone, mapping BPM, beat detection, and frequency bands to real-time image filtering on a VGA display.

My Contribution: VGA display controller, image rendering pipeline, parameterised 5×5 convolution filter with separable kernels (1px/cycle at 25MHz), weighted sum kernel scaling, and intensity/hue pixelwise filters. Image changing per beat (left and right paws swap)

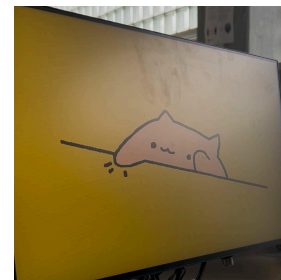
Result: System performed real-time audio-reactive image filtering, demonstrating complete FPGA data path integration from analog input to visual output. Received 100% on the project.



Original



Edge Detection



Hue Filter



QR Code for Video of Project

WHACK-A-MOLE FPGA GAME

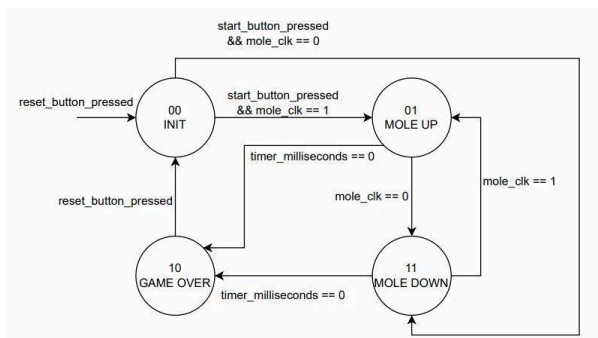
MTRX3700: Mechatronics 3 | FPGA Group Project

Verilog | FPGA (DE2-115) | FSM Design | Gray Coding | ModelSim

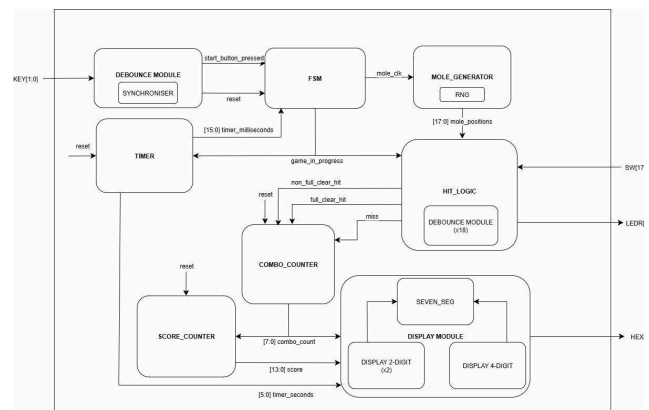
Designed and implemented a Whac-A-Mole game on an FPGA board (DE2-115) using Verilog. LEDs represent moles, toggle switches register hits/misses. All game logic, timing, and scoring handled entirely in hardware for low-latency, deterministic performance.

My Contribution: Designed the Finite State Machine (FSM). Developed a custom Moore-style FSM managing transitions between INIT, MOLE_UP, MOLE_DOWN, and GAMEOVER states. Gray-coded state encoding minimized transition errors. FSM coordinated all gameplay states, timing signals, and control outputs.

Result: Achieved stable gameplay with correct mole timing and score updates. Live demo performed seamlessly with responsive LED updates and scoring.



FSM State Transition Diagram



System Block Diagram



QR Code for Video of Project